



NORTH SYDNEY BOYS HIGH SCHOOL

MATHEMATICS EXTENSION 2

2021 HSC Course Assessment Task 1

13 December 2021

General instructions

- Working time – 1 hour. (plus 5 minutes reading time)
- Write your answers in the answer booklet provided.
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.
- Attempt **all** questions.

Student Number

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Class (please tick)

- ☐ 12MX2 A4 Mr Berry
- ☐ 12MX2 B6 Ms Lee
- ☐ 12MX2 C5 Mr Umakanthan

Marker's use only.

QUESTION	1	2	3	4	5	Total
MARKS	$\overline{6}$	$\overline{6}$	$\overline{7}$	$\overline{7}$	$\overline{9}$	$\overline{35}$



	MEX 12-4	MEX 12-2
Q1	/6	
Q2	/6	
Q3		/7
Q4	/7	
Q5	/9	
Total	/28 ²	/7

Question 1. (6 Marks)

- (a) Let $\omega = \frac{20 + 21i}{29}$ and $z = \frac{3 + 4i}{5}$
- (i) Express ωz in $(x + iy)$ form. 1
- (ii) Express $\omega \bar{z}$ in $(x + iy)$ form. 1
- (iii) Hence find two distinct ways of writing 145^2 as the sum $a^2 + b^2$ where a and b are integers such that $0 < a < b$ 1
- (b) In an Argand diagram, ABCD is a rhombus with $\angle ABC = 60^\circ$. 3
The points A and B represent the complex numbers $(3 + 4i)$ and $(6 + 6i)$ respectively.
Find the complex numbers representing the points C and D.
(your answer in $(a + ib)$ form with exact values for a and b .)

Question 2. (6 Marks)

- (i) Find the square roots of $8 + 6i$ 2
- (ii) Solve $z^2 + 2(2 + 3i)z - 13 + 6i = 0$ 2
- (iii) Describe geometrically the locus of z , if $|z + 5 + 4i| = |z^2 + 2(2 + 3i)z - 13 + 6i|$ 2

Question 3. (7 Marks)

- (a) Write down the negation to the statement "For all $x \in \mathbb{R}$, if $f(x) = \frac{2x + 3}{x + 2}$, then $f(x) \neq 2$ " 1
- (b) Write down the converse statement to the conditional statement "If $5x - 1 = 9$ then $x = 2$ " 1
- (c) Let a, b and n are integers. Prove by contrapositive the conditional statement: "If n does not divide ab then n does not divide a and n does not divide b " 2
- (d) Prove by contradiction that there are no rational solution to the equation $x^3 + x + 1 = 0$. 3

Question 4. (7 Marks)

Let $|z| \leq 2$ and $\arg\left(\frac{z+2}{\sqrt{3}+i}\right) \geq \frac{\pi}{12}$.

- (i) Sketch on an Argand diagram the region representing z . 3
- (ii) Hence determine the exact value of a and b such that $a \leq \arg(z+i) \leq b$ 2
- (iii) Find the minimum value of $|z+i|$ 2

Question 5. (9 Marks)

- (i) Solve the equation $z^5 + i = 0$. Write down all the roots in mod-arg form. 2
- (ii) Express $z^4 - iz^3 - z^2 + iz + 1$ as the product of two quadratic factors in z 2
- (iii) Hence evaluate $\left(\sin \frac{\pi}{10} \sin \frac{3\pi}{10}\right)$ 1
- (iv) Hence evaluate $\left(\sin \frac{3\pi}{10} - \sin \frac{\pi}{10}\right)$ 1
- (v) Hence find the exact value of $\sin\left(\frac{\pi}{10}\right)$ 3

End Of Paper

Suggested solution

Supplementary Task 1 Ext 2 2022

Q1 (a) "If $n^2 + 4n + 1$ is even, then n is odd"
contrapositive statement:

"If n is even then $n^2 + 4n + 1$ is odd."

(b) (i) "If $n^3 + 5$ is odd then n is even".

Negation:

both $n^3 + 5$ and n are odd.

Prove by negation (ii) Let $n = 2k + 1$ and $n^3 + 5 = 2j + 1$

$$2j + 1 = n^3 + 5$$

$$= (2k + 1)^3 + 5$$

$$2j + 1 = 8k^3 + 12k^2 + 6k + 6$$

$$\Rightarrow 2j = 8k^3 + 12k^2 + 6k + 5$$

Dividing by 2 and rearranging, we have

$$j - 4k^3 - 6k^2 - 3k = \frac{5}{2}$$

This, however is impossible: $\frac{5}{2}$ is a non-integer rational number, while $j - 4k^3 - 6k^2 - 3k$ is an integer by the closure properties for integers.

Therefore, it must be the case that our assumption that when $n^3 + 5$ is odd then n is odd is false, so, n must be even.

Proof by contrapositive

IF n is odd then $n^3 + 5$ is even.

$$\text{Let } n = 2k + 1 \quad k \in \mathbb{Z}.$$

$$n^3 + 5 = (2k + 1)^3 + 5$$

$$= 2[4k^3 + 6k^2 + 3k + 3]$$

$$= 2 \times \text{integer}$$

$$\therefore n^3 + 5 \text{ is even.}$$

Q2. (a) (i) $24 - 70i = (7 - 5i)^2$
 $\therefore \sqrt{24 - 70i} = \pm(7 - 5i) \checkmark$

or Standard Method

Let $\sqrt{24 - 70i} = a + ib$ where a, b are real.

$$\Rightarrow 24 - 70i = (a + ib)^2$$

$$= (a^2 - b^2) + 2iab$$

Matching the real and imaginary parts,

$$\left. \begin{array}{l} a^2 - b^2 = 24 \\ ab = -35 \end{array} \right\} \Rightarrow a^4 - 24a^2 - 35^2 = 0$$

$$(a^2 - 49)(a^2 + 25) = 0$$

$$\Rightarrow a = \pm 7 \quad a^2 + 25 \neq 0 \text{ since } a \text{ is real}$$

$$\left. \begin{array}{l} a = 7 \\ b = -5 \end{array} \right\} \quad \left. \begin{array}{l} a = -7 \\ b = 5 \end{array} \right\}$$

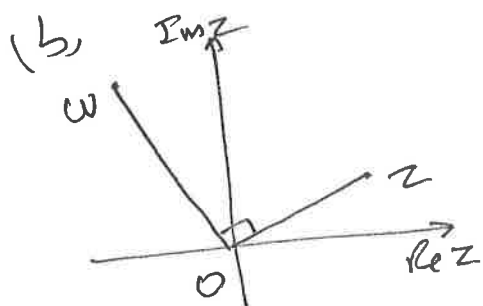
$$\therefore \sqrt{24 - 70i} = \pm(7 - 5i)$$

(ii) $z^2 - (5 - i)z + 15 - i = 0$
 $z = \frac{(5 - i) \pm \sqrt{(5 - i)^2 - 4 \times 15 - i}}{2}$

$$= \frac{(5 - i) \pm \sqrt{24 - 70i}}{2}$$

$$= \frac{(5 - i) \pm (7 - 5i)}{2}$$

$$= \underline{(6 - 3i)} \text{ or } \underline{(-1 + 2i)}$$



$$w = 2iz \checkmark$$

$$w^2 = (2iz)^2$$

$$= -4z^2 \checkmark$$

$$\Rightarrow \underline{w^2 + 4z^2 = 0}$$

Q3 (a) $z^5 + 1 = 0$

$$z^5 = -1$$

$$= \text{cis}(2k+1)\pi$$

$$z = \text{cis}(2k+1)\frac{\pi}{5}$$

$k=0$ $z_1 = \text{cis}\frac{\pi}{5}$

$k=-1$ $z_2 = \text{cis}(-\frac{\pi}{5})$

$k=1$ $z_3 = \text{cis}(\frac{3\pi}{5})$

$k=-2$ $z_4 = \text{cis}(-\frac{3\pi}{5})$

$k=2$ $z_5 = \text{cis}(\pi) = -1$

$$z^5 + 1 = (z+1)(z^4 - z^3 + z^2 - z + 1) = (z+1)[(z-z_1)(z-z_2)][(z-z_3)(z-z_4)]$$

$$\therefore z^4 - z^3 + z^2 - z + 1 = [z^2 - (z_1+z_2)z + z_1z_2][z^2 - (z_3+z_4)z + z_3z_4]$$

$$= [z^2 - (2\cos\frac{\pi}{5})z + 1][z^2 - (2\cos\frac{3\pi}{5})z + 1]$$

Matching the coefficients of z^3 :

$$-2\cos\frac{\pi}{5} - 2\cos\frac{3\pi}{5} = -1$$

$$\Rightarrow \cos\frac{\pi}{5} + \cos\frac{3\pi}{5} = \frac{1}{2}$$

Matching the coefficients of z^2 :

$$2 + 4\cos\frac{\pi}{5}\cos\frac{3\pi}{5} = 1$$

$$\Rightarrow \cos\frac{\pi}{5}\cos\frac{3\pi}{5} = -\frac{1}{4}$$

The equation whose roots are $\cos\frac{\pi}{5}$ and $\cos\frac{3\pi}{5}$ is

$$x^2 - \frac{1}{2}x - \frac{1}{4} = 0$$

$$4x^2 - 2x - 1 = 0 \Rightarrow x = \frac{1 \pm \sqrt{5}}{4}$$

as $\cos\frac{3\pi}{5} < 0$

$$\cos\frac{3\pi}{5} = \frac{1 - \sqrt{5}}{4}$$

